

Show understanding of the need for: assembler software, a compiler, an interpreter

9618_w25_qp_12_sc_10.a

QUESTION

10 A programmer is developing a computer program.

- (a) Explain how the programmer can first use an interpreter **and** then a compiler to develop the computer program.

Interpreter

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Compiler

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[4]

MARK SCHEME

10(a)	1 mark per bullet point, max 4 marks Max 2 from each section. Interpreter: • Use an interpreter while writing/coding the program • ... to test / debug the partially completed program • ... because errors can be corrected and processing continue from where the execution stopped // errors can be corrected in real time // errors are identified one at a time Compiler: • Use the compiler after the program is complete • ... to create an executable file so source code not seen • Use the compiler to repeatedly test the same completed section without having to re-interpret every time	4
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QUESTION

3 A programmer is writing a program in a high-level language.

(a) Complete the description of compilers and interpreters by writing the missing words.

A compiler checks all the code before attempting to translate the program. If any errors are found, they are all reported at the same time and the program does not translate or run. If there are no errors found, the compiler produces which can run without access to the

An interpreter translates one line of code and then runs it, before moving to the next line of code. If the line of code has an error, the interpreter and displays the error. The programmer can correct the error and then the interpreter continues translating from that point.

[4]

MARK SCHEME

3(a)	1 mark for each correctly completed term • an executable file/.exe • source/program code • stops • immediately/in real-time A compiler checks all of the code before attempting to translate the program. If any errors are found, they are all reported at the same time and the program does not translate or run. If there are no errors found, the compiler produces an executable file/.exe which can run without access to the source/program code. An interpreter translates one line of code and then runs it, before moving to the next line of code. If the line of code has an error, the interpreter stops and displays the error. The programmer can correct the error immediately/in real-time and then the interpreter continues translating from that point.	4
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QUESTION

(d) Francis's team use language translators.

Complete the descriptions of language translators by writing the missing words.

..... are usually used when a high-level language program is complete. They translate all the code at the same time and then run the program.

They produce files that can be run without the source code.

..... translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line.

Assemblers are used to translate assembly code into

[4]

MARK SCHEME

4(d)	1 mark for each correctly completed term Compilers are usually used when a high-level language program is complete. They translate all the code at the same time and then run the program. They produce executable/.exe/object code files that can be run without the source code. Interpreters translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line. Assemblers are used to translate assembly code into binary/machine code.	4
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QUESTION

- (b) Jennifer uses an Integrated Development Environment (IDE) to write her computer program.
- (i) The IDE allows Jennifer to use both an interpreter and a compiler while creating her computer program.

Describe the ways in which Jennifer can use **both** a compiler **and** an interpreter while developing the program.

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..... [4]

MARK SCHEME

7(b)(i)	1 mark per bullet point to max 4; max 3 from each section Interpreter: • Use an interpreter while writing the program • ... to test/debug the partially completed program • ... because errors can be corrected and processing continue from where the execution stopped // errors can be corrected in real time // errors are identified one at a time Compiler: • Use the compiler after the program is complete • ... to create an executable file • Use the compiler to repeatedly test the same (completed) section • ... without having to re-interpret every time // compiler not needed at run-time	4
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Explain the benefits and drawbacks of using either a compiler or interpreter and justify the use of each

9618_w25_qp_13_sc_8.a

QUESTION

8 Compilers and interpreters are used to translate programs written in a high-level language into a low-level language.

(a) State **two** disadvantages of using a compiler compared to an interpreter during program development.

1

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2

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[2]

MARK SCHEME

8(a)	1 mark per bullet point, max 2	2
	• Large amounts of source code take time to compile • It can be slower to produce the object code than an interpreter • The code must be recompiled when it is changed • The program cannot run if there are errors • It is not possible to correct errors in real-time • One error can cause false reporting of multiple further errors • Sections of code / unfinished code cannot easily be tested	

QUESTION

10 A programmer is developing a computer program.

- (a) Explain how the programmer can first use an interpreter **and** then a compiler to develop the computer program.

Interpreter

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.....

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Compiler

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.....

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[4]

MARK SCHEME

10(a)	1 mark per bullet point, max 4 marks Max 2 from each section. Interpreter: • Use an interpreter while writing/coding the program • ... to test / debug the partially completed program • ... because errors can be corrected and processing continue from where the execution stopped // errors can be corrected in real time // errors are identified one at a time Compiler: • Use the compiler after the program is complete • ... to create an executable file so source code not seen • Use the compiler to repeatedly test the same completed section without having to re-interpret every time	4
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An interpreter translates one line of code and then runs it, before moving to the next line of code. If the line of code has an error, the interpreter and displays the error. The programmer can correct the error and then the interpreter continues translating from that point.

[4]

MARK SCHEME

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QUESTION

- 8 A programmer uses an Integrated Development Environment (IDE) to write a computer program. The IDE has both a compiler and an interpreter as built-in translators.
- (a) The programmer decides to use the compiler when testing the final program.

Describe the benefits of using the compiler during testing.

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..... [2]

MARK SCHEME

8(a)	1 mark each to max 2: • Creates an executable file • ... so the code can be tested multiple times without having to recompile • ... so repeated testing takes less time	2
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QUESTION

- 3 A software developer is writing a computer program.
- (a) The developer uses an interpreter while writing the program code because it is easier for debugging.

Explain **one** reason why it is easier to debug the program code using an interpreter instead of a compiler.

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..... [2]

MARK SCHEME

3(a)	1 mark each to max 2: • The interpreter will stop when an error is found • ... so the error can be corrected in real-time, and the result of changes seen immediately • Only one error is displayed at a time • ... so fewer errors to correct simultaneously and no dependent errors	2
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9618_s24_qp_11_sc_3.b

QUESTION

(b) The program is ready to be sold to customers.

The developer uses a compiler because it creates an executable file.

Explain the reasons why the need to create an executable file makes the compiler the appropriate choice when the program is complete.

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..... [3]

MARK SCHEME

3(b)	1 mark each to max 3: • Program can be distributed without source code • ... so it cannot be edited/stolen/plagiarised • Users do not require the translator to run the program • ... so time is not spent retranslating by user	3
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QUESTION

6 A programmer uses both a compiler and an interpreter to translate a program written in a high-level language.

(a) Describe the advantages of using the interpreter compared to the compiler to translate the program.

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..... [4]

MARK SCHEME

6(a)	1 mark for each bullet point (max 4) • easier to debug the program • ... because it translates line-by-line and stops when an error is found whereas the compiler translates all the program at the same time • ... only reporting one error at a time • ... which allows the error to be corrected in real time whereas the program would need to be corrected and recompiled • ... and the program can restart at same point when error occurred with a compiler the program needs to be re-run • The effect of any changes made by the programmer can be seen immediately with a compiler the effects can only be seen after re-running • A partially completed program can be translated / tested on its own a compiler cannot translate a partial program	4
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QUESTION

- 5 A programmer is developing a computer game in a high-level language to sell to the public.
- (a) The programmer uses both an interpreter and a compiler at different stages of the development of the program.
- (i) Explain the reasons why the programmer uses an interpreter while writing the program code.

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..... [2]

MARK SCHEME

5(a)(i)	1 mark each to max 2 • Programmer can test sections of the code without every part working / being written • Programmer can debug in real time • ... so that errors can be fixed and the program continued from that point • The effect of any changes made by the programmer can be seen immediately • To avoid dependent errors	2
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QUESTION

- (ii) Explain the reasons why the programmer uses a compiler when the program has been written.

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..... [3]

MARK SCHEME

5(a)(ii)	1 mark each to max 3 • The compiler produces an executable file • ... so the user cannot access / edit / sell the code • ... and users do not need the translator to run the game • The game can be compiled for different hardware specifications • ... and then used to generate more income for the programmer • The program can be tested multiple times without having to retranslate each time	3
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9618_s23_qp_12_sc_7.b

QUESTION

- (b) The development team needs to use a translator whilst writing the program for the client.

Identify whether an interpreter or a compiler would be more appropriate at this stage of the program development.

Justify your choice.

Translator

Justification

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..... [3]

MARK SCHEME

7(b)	No mark for choice. 1 mark each to max 3 for justification Interpreter • Allows the developer to make real-time changes • ... so the program can be debugged at each stage • ... the effect of any changes made by the developer can be seen immediately • The developer can test when incomplete • ... so small parts can be tested without having to test the rest of the program • ... if one section does not work others can still be tested • To avoid dependent errors Compiler • The developer can debug multiple errors simultaneously • Produces an executable file • ... so that the developer can test the program multiple times without recompiling	3
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9618_w22_qp_12_sc_1.b.i

QUESTION

(b) Compilers and interpreters translate programs written in a high-level language into a low-level language.

(i) State **two** drawbacks of using a compiler compared to an interpreter during program development.

1

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2

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[2]

MARK SCHEME

1(b)(i)	1 mark for each bullet point (max 2): • larger amounts of source code take time to compile • slower to produce the object code than an interpreter • code cannot be changed without recompilation • the program will not run if there are any errors • errors cannot be corrected in real-time • one error may result in other false errors being reported • cannot easily test specific sections of the source code // cannot easily test unfinished source code	2
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QUESTION

- (b) Jennifer uses an Integrated Development Environment (IDE) to write her computer program.
- (i) The IDE allows Jennifer to use both an interpreter and a compiler while creating her computer program.

Describe the ways in which Jennifer can use **both** a compiler **and** an interpreter while developing the program.

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..... [4]

MARK SCHEME

7(b)(i)	1 mark per bullet point to max 4; max 3 from each section Interpreter: • Use an interpreter while writing the program • ... to test/debug the partially completed program • ... because errors can be corrected and processing continue from where the execution stopped // errors can be corrected in real time // errors are identified one at a time Compiler: • Use the compiler after the program is complete • ... to create an executable file • Use the compiler to repeatedly test the same (completed) section • ... without having to re-interpret every time // compiler not needed at run-time	4
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Show awareness that high-level language programs may be partially compiled and partially interpreted, such as Java (console mode)

9618_w23_qp_11_sc_6.b

QUESTION

(b) State **one** reason why some high-level languages are partially compiled and partially interpreted.

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..... [1]

MARK SCHEME

6(b)	1 mark for each bullet point (max 1) - Partially compiled programs can be used on different platforms as they are interpreted when run - Code is optimised for the CPU as machine code is generated at run time - Source code does not need recompiling so more efficient to run	1
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9618_w22_qp_12_sc_1.b.ii

QUESTION

(ii) Explain why high-level language programs might be partially compiled and partially interpreted.

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..... [2]

MARK SCHEME

1(b)(ii)	1 mark for each bullet point (max 2): - partially compiled programs can be used on different platforms as they are interpreted when run - code is optimised for the CPU as machine code is generated at run time	2
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9618_w25_qp_11_sc_5.c

[2]

2

..... [4]

MARK SCHEME

4(a)	1 mark for each bullet point (max 4) e.g. • Single stepping • Run the program one line at a time • ... and check the variable contents / program flow // show the effect of each line of code • Set breakpoints • ... run the code up to a set line • ... and then check the status • Variable/report watch window • ... view how the data changes as the program is running	4
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9618_w24_qp_11_sc_4.d.i

QUESTION

(d) The program code is written using an Integrated Development Environment (IDE).

(i) One presentation feature found in a typical IDE is prettyprint.

Identify **and** describe **one other** presentation feature found in a typical IDE.

Feature

Description

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[2]

MARK SCHEME

4(d)(i)	1 mark for correct name, 1 mark for matching description • Expand/collapse code blocks... • ... sections of source code that are part of the same block can be expanded to see the content or collapsed so that the overall code is seen • Auto-indentation // auto-formatting... • ... automatically indents/formats code as the user types so that the structure is clear // aids readability	2
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QUESTION

(ii) One debugging feature found in a typical IDE is single stepping.

Identify **and** describe **one other** debugging feature found in a typical IDE.

Feature

Description

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.....

[2]

MARK SCHEME

4(d)(ii)	1 mark for correct name, 1 mark for matching description • Breakpoints ... • ... stops the code running on a set line to view the current status/variable contents/program flow • Report window // variable watch window ... • ... shows the values in variables/data structures and how they change when each line is run	2
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QUESTION

(b) IDEs have many features other than built-in translators.

Complete the table by identifying **one other** common IDE feature that can be used for each purpose. Describe how each feature helps the user during program development.

Each feature must be different. Do **not** give translator as one of your features.

Purpose	IDE feature	Description
for coding		
for presentation		
for debugging		

[6]

MARK SCHEME

8(b)

1 mark for identification of each feature and 1 mark for matching description:

6

e.g.

For coding:

IDE feature	Description
Context-sensitive prompts	Gives suggestions for code as the user types instead of having to write/remember the code
Auto-correct	Corrects spelling mistakes so that user has fewer errors to correct

For presentation:

IDE feature	Description
Pretty-printing	Colour code keywords so the user can identify any errors
Expand/collapse (code) blocks	The user can hide code that they are not currently working on

For debugging:

IDE feature	Description
Single stepping	Run the code one line at a time // shows the effect of each line of code
Breakpoints	Stop the code running at a set point to check the flow/variable contents

QUESTION

- 5 A programmer uses an Integrated Development Environment (IDE) to develop a program that monitors air quality.

(a) Describe the following features of a typical IDE.

Context-sensitive prompts

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Single stepping

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[4]

MARK SCHEME

5(a)	1 mark for each bullet point (max 2 for each feature). Context-sensitive prompts: - As the code is being written ...the options to complete the statement are shown Single stepping: - allows the programmer to execute the program one line at a time ...so that the effects of each statement can be seen	4
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QUESTION

- (c) (i) Identify **two** features that support the visual presentation of the code in a typical Integrated Development Environment (IDE).

1

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2

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[2]

MARK SCHEME

6(c)(i)	1 mark for each bullet point (max 2) • Prettyprint • Expand/collapse code blocks • Auto indentation / formatting	2
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9618_w23_qp_11_sc_6.c.ii

QUESTION

(ii) Identify **two** features that support the debugging of the code in a typical IDE.

1

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2

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[2]

MARK SCHEME

6(c)(ii)	1 mark for each bullet point (max 2) • Single stepping • Breakpoints • Report window • Variable expressions	2
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QUESTION

(c) The development team uses an Integrated Development Environment (IDE).

Complete the table by describing the typical features found in an IDE.

Feature	Description
Breakpoints	
Dynamic syntax checks	
Context-sensitive prompts	
Single stepping	

[4]

MARK SCHEME

7(c)	1 mark each Breakpoints: • Stop the code at a specific line to check the current progress / values Dynamic syntax checks: • Highlight / underline / colour syntax errors as the code is entered Context-sensitive prompts: • Suggest the code to add // automatically complete statements Single stepping: • Run the code one line at a time so the values can be checked	4
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QUESTION

(ii) A typical IDE provides debugging tools to support the testing of a program.

Identify **three** other tools or features found in a typical IDE to support the writing of the program.

1

2

3

[3]

MARK SCHEME

4(b)(ii)	1 mark for each correct tool e.g. • Colour coding // pretty printing • Auto-complete • Auto-correct • Context sensitive prompts • Expand and collapse code blocks	3
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QUESTION

(ii) Identify **two** debugging tools that a typical IDE can provide.

1

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2

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[2]

MARK SCHEME

7(b)(ii)	1 mark per correct tool to max 2 e.g. • Breakpoints • Single stepping • Report windows	2
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